

Sobol' HDMR, Hoeffding HDMR and Shapley Values

Thierry A. Mara

University of Reunion

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Outline

Notations

Hoeffding HDMR

Rosenblatt Transformations

Sobol' HDMR

Shapley Value

Variance-based sensitivity indices

Conclusion

Example

Boston House Dataset

Notations & Assumptions

- ▶ $y = f(\mathbf{x}) + \varepsilon$ the scalar QoI
- ▶ $\mathbf{x} = (x_1, x_2, \dots, x_d) \sim p_{\mathbf{x}} = \frac{\partial^d F_{\mathbf{x}}}{\partial x_1 \partial \dots \partial x_d}$
- ▶ $x_i \sim p_i = \frac{dF_i}{dx_i}$, the marginal pdf (F_i cdf) of x_i
- ▶ $p_{i|j} = \frac{dF_{i|j}}{dx_i}$, the conditional pdf (cdf) of x_i over $x_j, \dots$
- ▶ $\mathcal{D} = \{1, \dots, d\}$, $\mathcal{D}_{-i} \cap \{i\} = \emptyset$, $\mathcal{D}_{+i} \cap \{i\} \neq \emptyset$
- ▶ $\mathbf{x} = (\mathbf{x}_{\alpha}, \mathbf{x}_{-\alpha})$ with $\mathbf{x}_{\alpha} \cap \mathbf{x}_{-\alpha} = \emptyset$
- ▶ $\alpha = \{i_1, \dots, i_k\} \subseteq \mathcal{D} \Leftrightarrow \mathbf{x}_{\alpha} = \mathbf{x}_{i_1, \dots, i_k} = (x_{i_1}, \dots, x_{i_k})$
- ▶ $f_{\alpha}(\mathbf{x}_{\alpha}) = \mathbb{E}[f|\mathbf{x}_{\alpha}] = \int_{\mathbb{R}^{d-|\alpha|}} f(\mathbf{x}) p_{-\alpha|\alpha} d\mathbf{x}_{-\alpha}$

Assumption: $\mathbb{E}[\ell^2] = \int_{\mathbb{R}^d} \ell^2(\mathbf{x}) p_{\mathbf{x}} d\mathbf{x} < \infty$

Hoeffding HDMR

Hoeffding's High-Dimensional Model Representation:

$$f(\mathbf{x}) = \sum_{\alpha \subseteq \mathcal{D}} f_{\alpha}^H(\mathbf{x}_{\alpha}) \quad (1)$$

with $f_0^H = \mathbb{E}[f(\mathbf{x})]$, and,

$$f_{\alpha}^H(\mathbf{x}_{\alpha}) = f_{\alpha}(\mathbf{x}_{\alpha}) - \sum_{\beta \subsetneq \alpha} f_{\beta}^H(\mathbf{x}_{\beta}) = \sum_{\beta \subseteq \alpha} (-1)^{|\alpha| - |\beta|} f_{\beta}(\mathbf{x}_{\beta}) \quad (2)$$

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As a consequence, Eq.(1) can be rewritten:

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As a result:

- ▶ Hoeffding's hdmr is unique (even when $p_{\mathbf{x}} \neq \prod_{i=1}^d p_i$)
- ▶ The $f_{\alpha}^H(\mathbf{x}_{\alpha})$'s are pairwise $\perp$ only if $p_{\mathbf{x}} = \prod_{i=1}^d p_i$

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Hoeffding HDMr is a "zero-sum" game:

$$f(x_1, x_2) = f_0^H + f_1^H(x_1) + f_2^H(x_2) + f_{12}^H(x_1, x_2) = f_{12}(x_1, x_2)$$

Rosenblatt Transformations

Transforms $\mathbf{x} \sim p_{\mathbf{x}}$ into $\mathbf{u} \sim \mathcal{U}(0, 1)^d$

$$\begin{cases} u_{i_1} = F_{i_1}(x_{i_1}) \\ u_{i_2} = F_{i_2|i_1}(x_{i_2}|x_{i_1}) \\ \vdots \\ u_{i_d} = F_{i_d|i_1, \dots, i_{d-1}}(x_{i_d}|\mathbf{x}_{-i_d}) \end{cases} \quad (4)$$

- ▶ the u -variables are independent of each other by definition
- ▶ RT is unique only if $p_{\mathbf{x}} = \prod_{i=1}^d p_i$
- ▶ RT requires the knowledge of the conditional cdf's

As a consequence it also turns $f(\mathbf{x})$ into $g(\mathbf{u})$: $f(\mathbf{x}) = g(\mathbf{u})$.

Sobol' HDMR

Sobol' hdmr is written as follows,

$$g(\mathbf{u}) = \sum_{\alpha \subseteq \mathcal{D}} g_{\alpha}(\mathbf{u}_{\alpha}) \quad (5)$$

with $g_0 = \mathbb{E}[g(\mathbf{u})]$, and, $\int_0^1 g_{\alpha}(\mathbf{u}_{\alpha}) d\mathbf{u}_{\alpha} = 0, \forall \alpha \neq \emptyset$

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As a result:

- ▶ **Sobol's hdmr is not unique as the RTs are not unique (unless $p_x = \prod_{i=1}^d p_i$)**
- ▶ **The g_{α} 's are pairwise $\perp$**

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Turning back to the original variables, we can write,

$$f(\mathbf{x}) = \sum_{\alpha \subseteq \mathcal{D}} f_{\alpha}^{\mathcal{S}}(\mathbf{x}_{\alpha}) \quad (6)$$

It is shown that $f_{\alpha}^{\mathcal{S}} = f_{\alpha}^H, \forall \alpha \subseteq \mathcal{D}$ if and only if $p_{\mathbf{x}} = \prod_{i=1}^d p_i$

Shapley Value

Shapley value $\Phi_i(\mathbf{x})$ have several formulations among which,

$$\Phi_i(\mathbf{x}) = \sum_{\alpha \subseteq \mathcal{D}_{+i}} \sum_{\beta \subseteq \alpha} (-1)^{|\alpha| - |\beta|} \frac{f_{\beta}(\mathbf{x}_{\beta})}{|\alpha|}$$

such that,

$$f(\mathbf{x}) = f_0^H + \sum_{i=1}^d \Phi_i(\mathbf{x})$$

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Let $\mathbf{x}^*$ be a given draw, $f(\mathbf{x}^*)$ the associated response

The Shapley value $\Phi_i(\mathbf{x}^*)$ is the fair contribution of x_i to $(f(\mathbf{x}^*) - f_0^H)$.

Fair = the mutual contributions (correlations+interactions) are equally shared.

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Shapley values allow for interpreting (black-box) Model responses.

Shapley Value & Hoeffding

We have,

$$f(\mathbf{x}) = f_0^H + \sum_{i=1}^d \Phi_i(\mathbf{x}) \quad (7)$$

with

$$\Phi_i(\mathbf{x}) = \sum_{\alpha \subseteq \mathcal{D}_{+i}} \sum_{\beta \subseteq \alpha} (-1)^{|\alpha| - |\beta|} \frac{f_\beta(\mathbf{x}_\beta)}{|\alpha|} \quad (8)$$

By comparing, Eq.(8) to Eq.(3), that is,

$$f_\alpha^H(\mathbf{x}_\alpha) = \sum_{\beta \subseteq \alpha} (-1)^{|\alpha| - |\beta|} f_\beta(\mathbf{x}_\beta)$$

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We can infer that

$$\Phi_i(\mathbf{x}) = \sum_{\alpha \subseteq \mathcal{D}_{+i}} \frac{f_{\alpha}^H(\mathbf{x}_{\alpha})}{|\alpha|} \quad (9)$$

We see that Hoeffding's hdmr is the key for interpreting (black-box) Model responses.

Shapley Value: Computational Issue

We have,

$$f(\mathbf{x}) = f_0^H + \sum_{i=1}^d \Phi_i(\mathbf{x})$$

with

$$\Phi_i(\mathbf{x}) = \sum_{\alpha \subseteq \mathcal{D}_{+i}} \frac{f_{\alpha}^H(\mathbf{x}_{\alpha})}{|\alpha|}$$

and

$$f_{\alpha}^H(\mathbf{x}_{\alpha}) = \sum_{\beta \subseteq \alpha} (-1)^{|\alpha| - |\beta|} f_{\beta}(\mathbf{x}_{\beta})$$

Computing Shapley Values requires the computation of the $2^d - 1$ f_{α} -functions from which one can infer the $f_{\alpha}^H(\mathbf{x}_{\alpha})$'s.

$f_{\alpha}(\mathbf{x}_{\alpha}) = \mathbb{E}[f(\mathbf{x})|\mathbf{x}_{\alpha}]$ can be estimated with any regression technique (that mitigates over/under fitting)

Shapley Value: Computational Issue

Suppose that Sobol' hdmr has been obtained for the following RT ordering $(i_1, \dots, i_d)$, it has been shown that (Mara & Tarantola 2012):

- ▶ $f_0 = \mathbb{E}[g(\mathbf{u})] = g_0$
- ▶ $f_{i_1}(\mathbf{x}_{i_1}) = \mathbb{E}[g(\mathbf{u})|u_{i_1}] = g_0 + g_{i_1}(u_{i_1})$
- ▶ $f_{i_1, i_2}(\mathbf{x}_{i_1}, \mathbf{x}_{i_2}) = \mathbb{E}[g(\mathbf{u})|u_{i_1}, u_{i_2}] = g_0 + g_{i_1}(u_{i_1}) + g_{i_2}(u_{i_2}) + g_{i_1, i_2}(u_{i_1}, u_{i_2})$
- ▶ $\vdots$

d of the f_α -functions can be obtained from one single Sobol' hdmr
The remainder can be inferred from the Sobol's hdmr of different RT-orderings

Variance-based sensitivity indices

Suppose that the RT is applied to $(x_{i_1}, \dots, x_{i_d})$, that is

$$\left\{ \begin{array}{l} \textcolor{red}{u}_{i_1} = F_{i_1}(x_{i_1}) \\ u_{i_2} = F_{i_2|i_1}(x_{i_2}|x_{i_1}) \\ \vdots \\ u_{i_{d-1}} = \dots \\ \textcolor{blue}{u}_{i_d} = F_{i_d|i_1, \dots, i_{d-1}}(x_{i_d}|\mathbf{x}_{-i_d}) \end{array} \right.$$

and recall that, $f(\mathbf{x}) = g(\mathbf{u})$,

Sobol' indices: (see Mara and Tarantola 2012)

Index	dependence effect	interaction effect
$S_{i_1}^{full} = \frac{\mathbb{V}[\mathbb{E}[g(\mathbf{u}) u_{i_1}]]}{\mathbb{V}[g(\mathbf{u})]}$	yes	no
$T_{i_1}^{full} = \frac{\mathbb{E}[\mathbb{V}[g(\mathbf{u}) \mathbf{u}_{\sim i_1}]]}{\mathbb{V}[g(\mathbf{u})]}$	yes	yes
$S_{i_d}^{ind} = \frac{\mathbb{V}[\mathbb{E}[g(\mathbf{u}) u_{i_d}]]}{\mathbb{V}[g(\mathbf{u})]}$	no	no
$T_{i_d}^{ind} = \frac{\mathbb{E}[\mathbb{V}[g(\mathbf{u}) \mathbf{u}_{\sim i_d}]]}{\mathbb{V}[g(\mathbf{u})]}$	no	yes

Shapley effects:

$$Sh_{i_1} = \sum_{\alpha \subseteq \mathcal{D} - i_1} \frac{|\alpha|! (d - 1 - |\alpha|)!}{d!} \left(S_{\alpha \cup i_1}^{full} - S_{\alpha}^{full} \right) \quad (10)$$

with $S_{\alpha}^{full} = \frac{\mathbb{V}[\mathbb{E}[g(\mathbf{u})|\mathbf{u}_{\alpha}]]}{\mathbb{V}[g(\mathbf{u})]}$ the Sobol' indices of the subset $\mathbf{u}_{\alpha}$

Conclusion

Assumption: $p_x = \prod_{i=1}^d p_i$

	Hoeffding	Sobol'
Conditional cdfs knowledge	-	-
Uniqueness	Yes	Yes
Orthogonality	Yes	Yes
Shapley Values Estimate	Yes	Yes
Cost in terms of regression	1	1

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Assumption: $p_x \neq \prod_{i=1}^d p_i$

	Hoeffding	Sobol'
Conditional cdfs knowledge	No	Yes
Uniqueness	Yes	No
Orthogonality	No	Yes
Shapley Values Estimate	Yes	Yes
Cost in terms of regression	$2^d - 1$	$d! / \left(\frac{d}{2}!\right)^2$

Some References

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- ▶ Shao Q et al. (2017). Comput. Meth. in Appl. Mech. and Engrg., 474–496
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Boston Housing Example

The Boston housing dataset ($\mathbf{X}, \mathbf{y}$)

$\mathbf{X}$: has $d = 13$ inputs (i.e. predictors), some of which are discrete variables

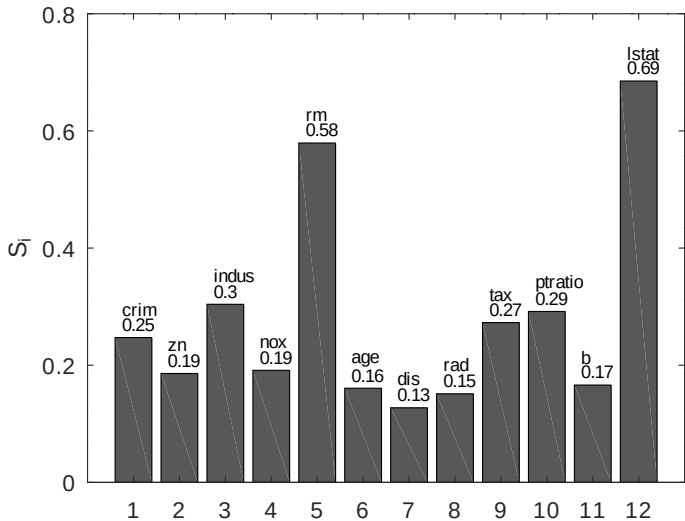
$\mathbf{y}$: the median value of the house

$N = 506$ data points

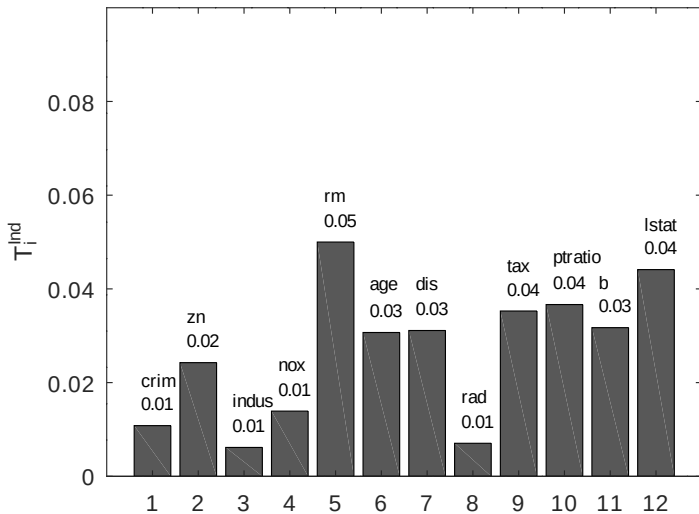
Assumption: $y = f(\mathbf{x}) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$

Sensitivity Analysis: Which inputs are the key drivers of the median value of the house?

Sobol' indices: S_i^{full}



Sobol' indices: T_i^{ind}



Shapley Effects

